

SAT Math Mastery · Trick Pack 1D

SAT Trick Pack 1 — Part D

Time 75 minutes: 55 min solving + 20 min self-check and error log

Topics exponential rates · angle bisectors · unit conversion · cylinders · confidence intervals · circles and tangency · parallel lines · spheres · quadratic intersections · altitude to hypotenuse · discriminants · proportional equations · pyramids · parametrised circles

Name / Date _____

1 · SKILLS & IDEAS

- S1. Read the per-unit growth factor of an exponential model and convert it to a percent change (and back from a multi-unit change).
- S2. Enumerate angle sums systematically from a small set of repeating angle types.
- S3. Chain unit conversions so units cancel; square the factor when the unit is squared.
- S4. Eliminate a dimension with a volume condition, solve the resulting cubic by testing integer roots, then apply the extra constraint.
- S5. Interpret a confidence interval and scale the margin of error with $1/\sqrt{n}$.
- S6. Use tangency (distance from center to line equals radius) to find a constant in a circle equation.
- S7. Force parallel lines with equal slopes, then confirm the intercepts differ.
- S8. Scale similar solids: radius ratio $\rightarrow$ volume ratio by cubing.
- S9. Write a quadratic as a line plus a multiple of the root factors when two graphs intersect.
- S10. Use the altitude to the hypotenuse: $h^2 = pq$, then choose the right root and a trig ratio.
- S11. Use the discriminant for “no intersection” and bracket a parameter between perfect squares.
- S12. Recognise proportional equations (infinitely many solutions) and scale constants with the coefficients.
- S13. Chain base $\rightarrow$ side $\rightarrow$ height $\rightarrow$ slant height $\rightarrow$ face area in a pyramid.
- S14. Test whether a parametrised point stays on a parametrised circle by simplifying its offsets from the center.

2 · REQUIRED FACTS

- If $f(x) = ab^{x/k}$, then $f(x+1)/f(x) = b^{1/k}$ and the percent change per unit is $b^{1/k} - 1$ (retained percent = $b^{1/k}$).
- A growth factor over n units is r^n ; the factor per unit is its n th root.
- A confidence interval is a range of plausible values, not certain ones; the margin of error is proportional to $\frac{1}{\sqrt{n}}$.
- A line is tangent to a circle exactly when the distance from the center to the line equals the radius.
- Two lines with equal slopes are parallel (no solution) if the intercepts differ and identical (infinitely many solutions) if they match.
- Spheres are similar: radii in ratio k give surface areas in ratio k^2 and volumes in ratio k^3 .
- Two graphs have equal outputs at their intersections, so the difference of the two functions has those x -values as roots.
- In a right triangle with altitude h to the hypotenuse splitting it into p and q : $h^2 = pq$.
- A system with no real solution leads to a quadratic with negative discriminant.
- Time appears squared in an acceleration; one length unit and two time units must be converted.

3 · REQUIRED FORMULAS

$$\text{percent change} = b^{1/k} - 1, \quad \frac{ME_2}{ME_1} = \sqrt{\frac{n_1}{n_2}}, \quad d = \frac{|Ah + Bk + C|}{\sqrt{A^2 + B^2}}, \quad V_{\text{cyl}} = \pi r^2 h, \quad S_{\text{cyl}} = 2\pi r^2 + 2\pi r h$$

$$\Delta = b^2 - 4ac, \quad V_{\text{pyr}} = \frac{1}{3}Bh, \quad \ell^2 = h^2 + \left(\frac{s}{2}\right)^2, \quad A_{\text{lat}} = 4 \cdot \frac{1}{2}s\ell, \quad S_{\text{sph}} = 4\pi r^2, \quad V_{\text{sph}} = \frac{4}{3}\pi r^3$$

4 · MINI THEORY SHEET

Per-unit factors. Compare outputs one input unit apart, or rewrite $b^{x/k} = (b^{1/k})^x$. Percent change = factor -1 ; do not report the retained percent.

Chained conversions. Write the rate as a fraction, multiply by conversion fractions so every unwanted unit cancels, then scale to the requested quantity. Squared units square the factor.

Lines and curves. If two graphs meet at $x = r, s$, then $p(x) = L(x) + a(x - r)(x - s)$; one more point fixes a .

Constraints on roots. A cubic from a volume condition usually has small integer roots; test divisors of the constant term, then use the stated inequality to pick the valid one.

Parametrised objects. Substitute the parametrised point and simplify; a parameter that cancels leaves a statement that is true or false for all values.

Speed shortcuts.

Situation	Shortcut
$f(x) = ab^{x/k}$	factor per unit is $b^{1/k}$; percent = factor -1
growth over n units	per-unit factor = (n th root of total factor)
rotating bisectors	list gaps ($\frac{x}{2}$ and $90 - \frac{x}{2}$), add consecutive ones
rate in mi/gal, want L for km	invert the rate, chain the factors
cylinder, volume and surface given	substitute $h = \frac{V}{\pi r^2}$, test small integer r
margin of error	$\propto \frac{1}{\sqrt{n}}$: four times the sample halves it
circle with tangent line	distance from center to line = r
$kx + 3y$ vs $4x + (k - 1)y$	equal slopes, then check the constants
sphere, circumference and area	radius ratio, then cube
parabola meets line at r, s	$p = L + a(x - r)(x - s)$, solve a from one point
altitude to hypotenuse	$h^2 = pq$; choose root by the inequality
parabola and line, no intersection	$\Delta < 0$; bracket between perfect squares
proportional equations	scale factor from coefficients, apply to the constant
pyramid	diagonal $\rightarrow$ side $\rightarrow$ height $\rightarrow$ slant $\rightarrow$ face
point on parametrised circle	simplify offsets; parameters must cancel
$\text{m/s}^2 \rightarrow \text{km/h}^2$	multiply by $3600^2/1000$

5 · GUIDED EXAMPLES

Example 1 — Squared time unit

A cart accelerates at 0.2 m/s^2 . Express this in km/h^2 .

$$0.2 \frac{\text{m}}{\text{s}^2} \cdot \frac{1 \text{ km}}{1000 \text{ m}} \cdot \left(\frac{3600 \text{ s}}{1 \text{ h}}\right)^2 = 0.2 \cdot \frac{12,960,000}{1000} = 2592 \text{ km/h}^2.$$

Example 2 — Line plus factored term

A quadratic p and the line $L(x) = x + 2$ intersect at $x = 1$ and $x = 5$, and $p(0) = -3$. Find $p(3)$.

$p(x) = x + 2 + a(x - 1)(x - 5)$. At 0: $2 + 5a = -3 \Rightarrow a = -1$. Then $p(3) = 5 - (2)(-2) = 9$.

Example 3 — No intersection means $\Delta < 0$

For which integers m do $y = x^2 + 2x + 5$ and $y = mx + 1$ have no common point?

$x^2 + (2 - m)x + 4 = 0$, $\Delta = (2 - m)^2 - 16 < 0 \Rightarrow |m - 2| < 4 \Rightarrow -2 < m < 6$. Integers: $-1, 0, \dots, 5$ (and $m = 6$ gives a tangent line, not allowed).

6 · PRACTICE SET — 16 PROBLEMS · ALL SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · CALCULATOR

The value V , in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18,000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

- (A) It decreases by 24.5% each year. (B) It decreases by 30% each year.
(C) It decreases by 70% each year. (D) It decreases by 51% each year.

2. SAT CHALLENGE · NO CALCULATOR NEEDED

Two lines intersect at point O , and the smallest of the four angles they form measures 50° . A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O , which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

- (A) 130° (B) 155°
(C) 140° (D) 115°

3.

SAT CHALLENGE · CALCULATOR

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

4.

SAT CHALLENGE · CALCULATOR

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 12

(B) 2

(C) 3

(D) 4

5.

SAT CHALLENGE · CALCULATOR

A researcher surveys a random sample of 625 workers in a city and calculates a 95% confidence interval of 24.2 to 27.8 minutes for the mean one-way commute time of all workers in the city. Assume a second random sample of 2,500 workers has similar variability and the same method is used. Which of the following statements must be true?

(A) About 95% of sampled commutes lie in the interval. (B) The new margin is about 0.45 min.

(C) The population mean cannot be 28.1 min. (D) The new margin is about 0.9 min.

6.

SAT CHALLENGE · CALCULATOR

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 - 6x + 8y = k$$

In the xy -plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x - 3y = -11$. What is the value of k ?

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$kx + 3y = 12$$

$$4x + (k - 1)y = 12$$

In the system of equations above, k is a constant. For which value of k does the system have no solution?

(A) 3

(B) -4 (C) -3

(D) 4

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A . What is the ratio of the volume of sphere B to the volume of sphere A ?

(A) $\frac{9}{4}$

(B) $\frac{27}{8}$

(C) $\frac{5}{4}$

(D) $\frac{3}{2}$

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x -coordinate 1 and one with x -coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

11.

SAT CHALLENGE · CALCULATOR

In right triangle ABC , the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?

(A) $\frac{\sqrt{5}}{5}$

(B) $\frac{4}{5}$

(C) $\frac{1}{2}$

(D) $\frac{2\sqrt{5}}{5}$

12.

SAT CHALLENGE · DESMOS

In the xy -plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m ?

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$6x - 2(y - 5) = 2x + 3y + 1$$

$$ax + 20y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $\frac{b}{a}$?

14.

SAT CHALLENGE · CALCULATOR

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 260

(B) 240

(C) 130

(D) 360

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x - a - 2)^2 + (y - 2a + 1)^2 = 25$$

In the xy -plane, the circle above is defined for each real number a . Which of the following points lies on the circle for every real value of a ?

(A) $(a - 1, 2a + 2)$ (B) $(a + 5, 2a + 3)$ (C) $(a + 5, 2a + 4)$ (D) $(2a + 5, 2a + 3)$

16.

SAT CHALLENGE · CALCULATOR

A commuter train accelerates at a constant rate of 0.5 meters per second squared. What is this acceleration, in kilometers per hour squared? (1 kilometer = 1000 meters and 1 hour = 3600 seconds.)